

Superconducting qubit readout: an open quantum system perspective

Kacper Kłos

30 września 2026

We start with system Hamiltonian for the transmon interacting with the resonator is (Formula (31) in Blase, ignoring the coupled transmission line)

$$\hat{H} = 4E_C(\hat{n} + \hat{n}_r)^2 - E_J \cos \varphi \quad (1)$$

Because $\hat{n} = \frac{\hat{Q}}{2e}$, $\hat{\varphi} = \frac{2e}{\hbar}\hat{\Phi}$, and $[\hat{\Phi}, \hat{Q}] = i$ that means

$$[\hat{n}, \hat{\varphi}] = \frac{1}{\hbar}[\hat{Q}, \hat{\Phi}] = -i \quad (2)$$

This allow us to see how $e^{i\hat{\varphi}}$ works on state $|n\rangle$ to compute the $\cos \hat{\varphi} |n\rangle$. We start with Campbell lemma:

$$e^{i\hat{\varphi}} \hat{n} e^{-i\hat{\varphi}} = \hat{n} + [i\hat{\varphi}, \hat{n}] + \frac{1}{2}[i\hat{\varphi}, [i\hat{\varphi}, \hat{n}]] = \hat{n} - 1 \quad (3)$$

This leads to:

$$\hat{n}(e^{-i\hat{\varphi}} |n\rangle) = (n - 1)e^{-i\hat{\varphi}} |n\rangle \quad (4)$$

So, the state $(e^{i\hat{\varphi}} |n\rangle)$ is the eigenstate with the eigenvalue $n + 2$. This leads to $(e^{\pm i\hat{\varphi}} |n\rangle) = |n \pm 1\rangle$. This is relevant for the $\cos \varphi = \frac{1}{2}(e^{i\hat{\varphi}} + e^{-i\hat{\varphi}})$. Thus, we know how the H_t that is H when $\hat{n}_r = 0$ work on the state $|n\rangle$, allowing us to describe the matrix.

$$\hat{H}_t |n\rangle = 4E_C n^2 |n\rangle - \frac{E_J}{2}(|n + 1\rangle + |n - 1\rangle) \quad (5)$$